F-loop — ε-free PSR/NSR reliably recover an interior θ* on a sloppy IQS valley; finite differences need an ε they cannot tune without the answer: too-large ε FAILS (truncation $\frac{\varepsilon^2}{6}|C'''|/\mu_{\mathrm{soft}}$), too-small ε is UNRELIABLE (δ/ε noise, wide IQR), and even the best ε only reaches an uncertifiable δ-floor
(2q TFIM, $P=2$; compiled to machine-native segments, emulated under T4; $T/T_2^*=0.15$; 20 seeds, median ± IQR; $w=0.250$, $\mu_{\mathrm{soft}}=0.54$)